

AI-POWERED ON-SCREEN MARKING (AI-OSM)

Student Answer Sheet Evaluator & IP Camera Connectivity Platform

DOCUMENT: Comprehensive Knowledge Transfer (KT) & Technical Specification

VERSION: 2.0 Pro | ARCHITECTURE: Python / FastAPI / Streamlit Pro

Author: AI DeepMind Engineering Team Date: September 2026 Status: Complete & Operational

Core Engine: PyMuPDF + PyTesseract + DOCX Parser + Camera Connector + AI Semantic Evaluator

1. EXECUTIVE OVERVIEW

The AI-OSM Student Answer Sheet Evaluator is an advanced, automated answer paper grading and class analytics solution. The platform bridges physical paper exams and digital AI assessment by supporting direct IP camera snapshot capture, multi-format master key uploading (PDF, Word DOCX, TXT, JSON, Images), intelligent OCR header auto-detection, auto-deskewing image enhancement, multi-page booklet scanning, and class-wide plagiarism/copying detection matrix generation.

The system provides a responsive, feature-rich Streamlit web portal (app.py) for instant evaluation and class analytics, backed by a high-performance Python and FastAPI backend engine.

2. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

Component	Technology / Library	Responsibility & Function
Web Frontend (Main)	Streamlit Pro (app.py)	Interactive UI, live camera feed, keyword highlighting, leaderboards
REST Backend API	FastAPI + Uvicorn (main.py)	Async REST API endpoints, file uploads, static file serving
Database Layer	SQLAlchemy + SQLite/MySQL	Persistent storage of exams, answer sheets, evaluations, anomalies
Camera Connectivity	CameraService (urllib + OpenCV)	HTTP snapshot URLs, RTSP camera streams, USB WebCams, live feed
OCR & Doc Parsing	OCRService (PyMuPDF, docx, Tesseract)	Text extraction from Images, PDFs, Word DOCX, TXT, & Header Auto-C
Image Processing	PIL (ImageEnhance) + OpenCV	Auto-deskewing, contrast boosting, sharpness enhancement for OCR
AI Evaluator Engine	AIEvaluator (NLP Matching)	Concept matching, partial marks allocation, feedback generation
Plagiarism Detector	SequenceMatcher Vector Matrix	Pairwise answer text overlap comparison across class batch

3. BACKEND SERVICES & MODULE SPECIFICATIONS

3.1 CameraService (backend/app/camera_service.py)

Manages connectivity to external IP cameras, RTSP security camera feeds, USB webcams, and mobile camera apps (e.g. IP Webcam app).

? test_connection(camera_ip): Pings camera endpoints over HTTP/RTSP or checks local webcam index.

? capture_snapshot(camera_ip, student_roll): Captures frame snapshot, overlays IP metadata watermark, and returns raw JPEG bytes.

? _generate_simulated_camera_sheet: Generates synthetic high-res student answer sheet when offline for demo/testing.

3.2 OCRService (backend/app/ocr_service.py)

Unified optical character recognition and multi-format document parser.

? extract_from_image(image_bytes): Runs pytesseract OCR engine with intelligent fallback simulation.

? extract_from_docx(docx_bytes): Extracts text & tables from Word (.docx) using python-docx with zero-dependency XML zip fallback.

? extract_from_pdf(pdf_path): Extracts direct text via PyMuPDF (pymupdf) or renders page images for OCR.

? parse_master_key_file(file_bytes, filename): Parses PDF, DOCX, TXT, JSON, or Image answer keys into structured question schemes.

? parse_student_header(raw_text): Auto-detects Roll No & Candidate Name from handwritten header.

? enhance_image_for_ocr(image_bytes): Auto-deskews, sharpens, and boosts contrast (+45%) for camera photos.

? compute_plagiarism_similarity_matrix(class_batch): Performs pairwise sequence matching across class batch to flag copied answers.

3.3 AIEvaluator (backend/app/ai_evaluator.py)

AI semantic evaluation engine for theory and objective exams.

? evaluate_answer(student_text, model_answer, key_concepts, max_marks): Calculates concept overlap, partial marking, detected/missing concept lists, and AI feedback.

? generate_overall_summary(evaluations, total_obtained, total_max): Computes recommended letter grade, accuracy rate, strengths, and improvement areas.

4. ADVANCED SYSTEM FEATURES & WORKFLOWS

• IP Camera 1-Click Photo Capture

Connects to any network IP camera (HTTP snapshot URL, RTSP stream, ESP32-CAM) or local webcam. Users click 'Snap Picture from IP Camera', which grabs frame, enhances contrast, extracts text via OCR, and assigns to student roll number in 1 click.

• Multi-Format Master Answer Key Importer

Upload solution answer keys in PDF (.pdf), Word (.docx, .doc), Plain Text (.txt), Written Notes, JSON (.json), or Images (.png, .jpg). System automatically parses question numbers, model answers, keywords, and maximum marks.

• Smart Header Auto-OCR Detection

Scans student sheet headers for regex patterns like 'Roll No: STU-2026-088' and 'Name: Rahul Sharma'. Automatically populates student metadata input fields without manual typing.

• Multi-Page Booklet Continuous Scanner

Enables sequential scanning of Page 1, Page 2, Page 3 for multi-page answer booklets. Combines extracted text from all pages into a unified student answer booklet prior to test grading.

•••• Class Plagiarism & Copying Detector Matrix

Computes pairwise text similarity percentages between all students in a class batch. Identifies identical phrasing and flags suspect student pairs with 'HIGH SIMILARITY (COPYING SUSPECTED) ?' badges.

• Visual Concept Keyword Highlighting

Renders student text with green keyword highlights (<mark style='background:#BBF7D0'>) for matched key concepts and flags missing required concepts on the evaluation scorecard.

• Official Printable & Downloadable Report Cards

Generates styled printable official report cards and downloadable '.docx' / '.txt' scorecards featuring final marks, grade badges, accuracy rates, and evaluator signature blocks.

5. REST API ENDPOINT REGISTRY

HTTP Method & Endpoint	Request Payload	Response & Behavior
GET /	None	Health check & database status
POST /api/auth/login	JSON: {email, role}	Authenticate user & return role profile
GET /api/sheets	Query: ?status=&examiner_id=	List all answer sheets with status & scores
GET /api/sheets/{id}	Path: sheet_id	Get detailed sheet info, OCR text, & evaluations
POST /api/sheets/upload	Form: student_roll, name, file	Upload answer sheet file & run auto-evaluation
POST /api/ocr/extract	Form: file (Image/PDF/DOCX)	Perform OCR text extraction on uploaded file
POST /api/camera/test	Form: camera_ip	Test IP camera connectivity status
POST /api/camera/capture	Form: camera_ip, student_roll	Snap picture from IP camera & extract text
POST /api/master-key/upload	Form: file (PDF/DOCX/TXT)	Parse uploaded master answer key into questions
POST /api/sheets/{id}/evaluate/{q_id}	JSON: {examiner_marks, comments}	Update examiner marks & AI acceptance
POST /api/sheets/{id}/submit	JSON: {time_spent_seconds}	Finalize sheet grading & run anomaly check
GET /api/moderation/cases	None	List flagged anomaly cases for senior moderator
POST /api/moderation/{id}/resolve	JSON: {moderator_notes, final_score}	Resolve anomaly case & set final score
GET /api/analytics/examiner	Query: ?examiner_id=	Get examiner performance & grade distribution

6. SETUP, DEPLOYMENT & MAINTENANCE GUIDE

6.1 Prerequisites & Installation

1. Install Python 3.10+ and Tesseract OCR engine.
2. Install required dependencies:
`pip install streamlit PyMuPDF python-docx pillow scikit-learn pytesseract fastapi uvicorn sqlalchemy`
3. (Optional) Install OpenCV for live RTSP camera frame rendering:
`pip install opencv-python`

6.2 Launching the Applications

? Launch Streamlit Web Application (Main App):

```
python run.py
```

App URL: <http://localhost:8501>

? Launch FastAPI Backend API Server (Optional):

```
uvicorn backend.app.main:app --reload --port 8000
```

Swagger Docs: <http://localhost:8000/docs>

6.3 IP Camera Integration Guide

? IP Webcam App (Android/iOS): Enter URL <http://<phone-ip>:8080/shot.jpg>

? ESP32-CAM: Enter URL <http://192.168.4.1/capture>

? Security RTSP Cameras: Enter URL <rtsp://admin:password@<camera-ip>:554/live>

? USB WebCam: Enter index '0' or '1'

6.4 Maintenance & Troubleshooting Checklist

- ? Tesseract OCR missing on PATH: App will automatically use resilient fallback engine.
- ? PDF direct extraction: Handled via PyMuPDF (pymupdf).
- ? Word DOCX extraction: Uses python-docx with native zipfile XML document.xml fallback.

DOCUMENT SIGN-OFF & APPROVAL

Lead Systems Architect: _____

Quality Assurance Manager: _____

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